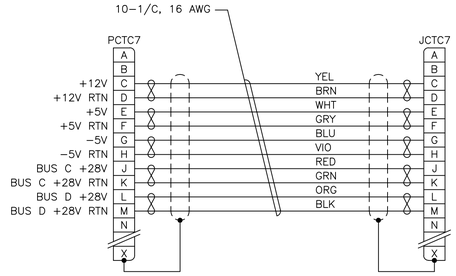
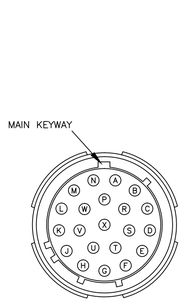
COMMERCIAL PARTS LIST

ITEM	QTY	PART NUMBER	DESCRIPTION	MANUFACTURER
C1	1	D38999/26FH21PB	CONNECTOR, CIRCULAR, 21 PINS	QPL
C2	2	380HS115UC2312A3	BACKSHELL, STRAIGHT	GLENAIR
C3	A/R	M39029/58-364	CONTACT, PIN, SIZE 16, 16-20 AWG	QPL
C4	1	D38999/20FH21SB	CONNECTOR, CIRCULAR, 21 SOCKETS	QPL
C5	A/R	M39029/56-352	CONTACT, SOCKET, SIZE 16, 16-20 AWG	QPL
C6	A/R	M22759/11-16-4	1/C, 16 AWG, PTFE INSULATED, YELLOW	QPL
C7	A/R	M22759/11-16-1	1/C, 16 AWG, PTFE INSULATED, BROWN	QPL
C8	A/R	M22759/11-16-9	1/C, 16 AWG, PTFE INSULATED, WHITE	QPL
C9	A/R	M22759/11-16-8	1/C, 16 AWG, PTFE INSULATED, GRAY	QPL
C10	A/R	M22759/11-16-6	1/C, 16 AWG, PTFE INSULATED, BLUE	QPL
C11	A/R	M22759/11-16-7	1/C, 16 AWG, PTFE INSULATED, VIOLET	QPL
C12	A/R	M22759/11-16-2	1/C, 16 AWG, PTFE INSULATED, RED	QPL
C13	A/R	M22759/11-16-5	1/C, 16 AWG, PTFE INSULATED, GREEN	QPL
C14	A/R	M22759/11-16-3	1/C, 16 AWG, PTFE INSULATED, ORANGE	QPL
C15	A/R	M22759/11-16-0	1/C, 16 AWG, PTFE INSULATED, BLACK	QPL
C16	–	–	–DELETED–	–
C17	A/R	THFE4-500015	HEAT SHRINK TUBING, EXPANDED PREHEAT ID 0.500"	SAINT-GOBAIN
C18	A/R	2146	BRAID, TINNED COPPER, 1/2"	ALPHA
C19	A/R	5011001332S	EXPANDABLE SLEEVING, 1/2", BLACK	BENTLEY HARRIS
C20	A/R	TZE-211	LABEL, P-TOUCH, 1/4", BLACK ON WHITE	BROTHER

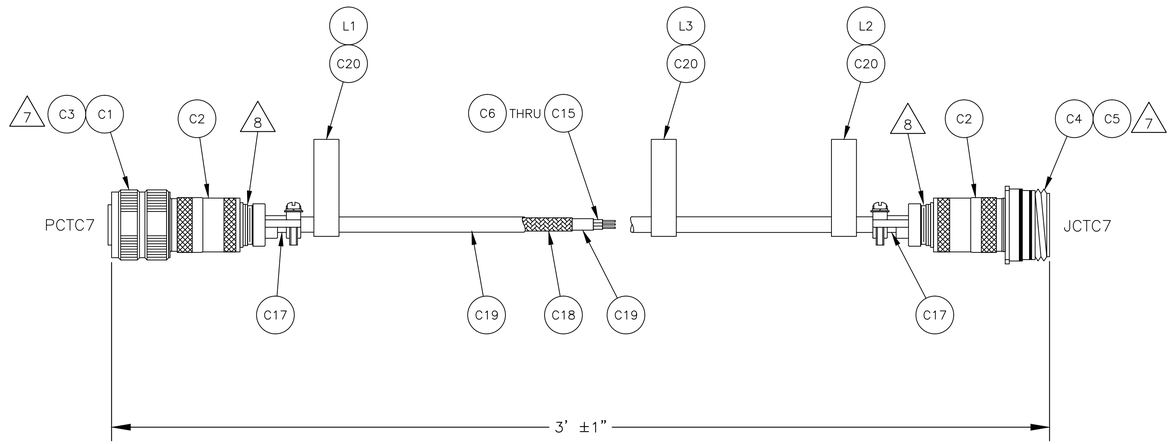
LABEL LIST	
ITEM	INSCRIPTION
L1	PCTC7
L2	JCTC7
L3	S1101EGA4567 SN xxxx

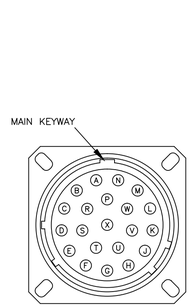


SCHEMATIC



CONNECTOR KEYING





CONNECTOR KEYING

NOTES:

1. MATERIAL CERTIFICATION REQUIRED PER QUALITY LEVEL SPECIFIED IN WORK ORDER/SOW.
2. ESD HANDLING AND CONTROL PER ESTABLISHED PROCEDURES.
3. HANDLE ALL PARTS AND ASSEMBLIES WITH CLEAN, LINT-FREE, ANTI-STATIC GLOVES OR FINGER COTS.
4. HARNESS ASSEMBLY AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.4 AND THE FOLLOWING NOTES:
   A. WIRE TWIST SHALL BE BETWEEN 8 AND 16 TIMES THE OUTER DIAMETER OF THE HARNESS UNLESS OTHERWISE NOTED. (FROM NASA-STD-8739.4 PARA 7.3.20)
5. FOR TENSILE PULLS SEE TENSILE PULL STRENGTH LIST. TENSILE PULL STRENGTHS PER NASA-STD-8739.4 AND/OR THE MANUFACTURER'S SPECIFICATIONS.
6. INSTALL CONTACTS IN ALL UNUSED POSITIONS OF CIRCULAR CONNECTORS.
7. PERFORM CONTACT RETENTION TEST PER NASA-STD-8739.4 AND/OR THE MANUFACTURER'S SPECIFICATIONS. CONTACTS REQUIRING A PULL TEST FOR RETENTION SHALL BE PERFORMED AS FOLLOWS: A GENTLE PULL ON THE WIRE SHALL BE USED TO DETERMINE IF THE CONTACT IS SEATED PROPERLY.
8. TORQUE BACKSHELLS (ITEM C2) TO 80 ±5 IN-LBS.
9. TEST PER NASA-STD-8739.4 PARA 18.2.6 USING THE FOLLOWING:
   CONTINUITY: < 5 OHMS
   DWV: VOLTAGE: 1500 ±75VDC
   CURRENT: < 1 MILLIAMP
   DURATION: < 1 MINUTE
   IR: VOLTAGE: 500 ±75VDC
   RESISTANCE: < 100 MEG OHMS
   DURATION: < 1 MINUTE
10. PLACE COMPLETED ASSEMBLY IN CLEAN, STATIC-SHIELDING PACKAGING. IDENTIFY PART BY TAGGING WITH PART NUMBER AND SERIAL/LOT NUMBER AS SPECIFIED IN WORK ORDER/STATEMENT OF WORK PER SAE AS478, METHOD 35D. STORE IN A PROJECT APPROVED STORAGE AREA.

CRIMP TOOL LIST				
CONTACT P/N	CRIMP TOOL	POSITIONER	I/E TOOL	REF. DES.
M39029/58-364	M22520/1-01	M22520/1-04	M81969/14-03	PCTC7
M39029/56-352	M22520/1-01	M22520/1-04	M81969/14-03	JCTC7

TENSILE PULL STRENGTH LIST			
MANUFACTURER	CONTACT P/N	WIRE AWG	LBS
QPL	M39029/58-364	16	41.0
QPL	M39029/56-352	16	41.0

BACKSHELL FASTENER TORQUE TABLE		
SCREW SIZE	TORQUE RANGE	UNITS
2-56	2 ±0.5	IN-LB
4-40	4 ±0.5	IN-LB
6-32	6 ±1.0	IN-LB
8-32	8 ±1.0	IN-LB
10-32	10 ±1.0	IN-LB
.250-20	12 ±1.0	IN-LB

REV	DESCRIPTION	APPROVED
A	REVISED PER ACME-ECO-401.	MH 2/4/15 TG 2/11/15 DS 3/11/15

REVISION HISTORY

NEXT ASSEMBLY

UNLESS OTHERWISE SPECIFIED:
1. ALL DIMENSIONS ARE IN INCHES
2. BREAK EDGES .010-.020
3. INSIDE CORNERS R.02 MAX
4. LINEAR TOLERANCES:
   .X ± 0.03
   .XX ± 0.01
   .XXX ± 0.005
5. ANGULAR TOLERANCE ± 0.5°

THIRD ANGLE PROJECTION

THIS DRAWING IS UNDER CONFIGURATION CONTROL: DO NOT REVISE WITHOUT AN APPROVED ECO.

APPROVED: E. McCLOSKEY 3/12/15
PRIOR TO USE, CHECK WITH CONFIGURATION MANAGEMENT FOR ANY OUTSTANDING ECOS.

DESIGNER	M. HOMKO 9/23/14	QUALITY	D. SEELEY 9/23/14
REVIEWER	J. BRUEWER 9/24/14	MATERIALS	M. WILKINSON 9/23/14
MECHANICAL	A. DRAKE 9/26/14	PROJ LEAD	B. BOROWSKI 9/29/14
ELECTRICAL	T. GOBELI 9/23/14		
STRUCTURAL	C. WERNER 9/24/14		
SAFETY	W. BORRELLI 9/24/14		

DIMENSIONS AND TOLERANCING PER ASME Y14.5M-1994

ZIN TECHNOLOGIES
6745 ENGLE ROAD
AIRPORT EXECUTIVE PARK
CLEVELAND, OHIO 44130
(440) 826-2200
www.zin-tech.com

CONTRACT NUMBER: NNC14CA02C
PROJECT CODE: 52-BT3

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
JOHN H. GLENN RESEARCH CENTER
LEWIS FIELD
CLEVELAND, OHIO

TITLE

PCTC7 EXTENSION CABLE (EMS SIDE SIMULATION)

SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101EGA4567	A

SCALE NONE | INITIAL REL DATE 09/29/14 | SHEET 1 OF 1

RELEASE LEVEL: RELEASED
MODEL NAME: S1101EGA4567_REV_A.DWG

AutoCAD 2012

DWG NO S1101EGA4567 | SH 1 | REV A